

Uncertainty-Aware Semantic Split Computing for Seizure Monitoring

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The Core Problem

- Continuous EEG streaming consumes high power
- High bandwidth usage
- Edge devices cannot run full deep models
- Cloud-only systems ignore sensing cost

Current systems optimize accuracy, not energy.

What if the device sensed more only when necessary?

Existing Approaches

- **Continuous raw EEG transmission**
- **Fixed sampling rate**
- **No adaptive sensing**
- **Either:**
 - Edge-only (low accuracy)
 - Cloud-only (high transmission cost)
- **No feedback loop**

Raw Signal Communication

- **Sends full waveform data**
- **Large packets**
- **High communication energy**
- **Redundant information**
- **Battery drains quickly**

Semantic Token-Based Communication

- Early CNN layers extract features
- Wearable sends compact semantic tokens
- Cloud processes meaningful representation
- Reduced transmission size
- Lower energy consumption
- Enables adaptive sensing

We transmit meaning, not raw signals.

Direct Comparison

Feature	Traditional	Our System
Data Sent	Raw EEG	Semantic Tokens
Sampling	Fixed	Adaptive
Intelligence	Static	Uncertainty-aware
Battery Life	Low	Optimized
Feedback Loop	None	Yes

The Semantic Feedback Loop

1. Wearable extracts semantic token
2. Cloud predicts seizure
3. Cloud computes uncertainty
4. If confidence low → request richer data
5. Device increases sensing temporarily
6. Returns to low-power mode

AI confidence controls hardware behavior.

What Makes It Novel

Not novel individually:

- CNN seizure detection
- Split inference
- Feature compression

Novel combination:

- Uncertainty-driven sensing
- Closed control loop
- Energy-aware AI system
- Semantic communication in biomedical monitoring

Three-Tier Architecture

Edge (Wearable)

- EEG sensor
- Lightweight CNN layers
- Token generation
- BLE/Wi-Fi transmission

Gateway (Raspberry Pi)

- Token buffering
- Device authentication
- Reliable forwarding
- Command relay

Cloud

- Deep CNN layers
- Uncertainty estimation
- Decision engine
- Alert system

Hardware Overview

Edge

- ESP32 / STM32
- Low-power modes
- Adaptive sampling

Gateway

- Raspberry Pi
- Local cache

Cloud

- GPU/CPU server
- Dashboard for doctors

Where Is the Patient?

- Patient wears device at home
- Gateway located in patient's home
- Cloud hosted at hospital/data center
- Doctors monitor remotely
- Real-time alerts to caregivers

Enables hospital-level monitoring at home.

Development Strategy

Phase 1: Cloud-Only

- Train CNN
- Implement uncertainty module
- Simulate token splitting

Phase 2: Gateway Simulation

- Token forwarding
- Command routing

Phase 3: Edge Deployment

- Deploy encoder
- Enable adaptive sensing

Advantages

- Extended battery life
- Reduced bandwidth
- Intelligent sensing
- Higher reliability
- Scalable to other bio-signals

Challenges

- Network dependency
- Tethered Range
- Slight latency
- System complexity
- Data privacy concerns
- Hardware constraints

Impact

- Remote seizure monitoring
- Telemedicine
- Rural healthcare
- Reduced hospital burden

Future Scope

- Federated learning
- Personalized models
- Cardiac monitoring
- Smart medical IoT ecosystem

Dataset Used for System Validation

Bonn EEG Dataset

- Public benchmark dataset
- Developed by University of Bonn, Germany
- Published by Andrzejak et al. (2001)
- Widely used for seizure detection research

Key Characteristics

- 5 subsets
- 100 EEG segments per subset
- Each segment: 23.6 seconds
- Sampling rate: 173.61 Hz
- Single-channel EEG

Bonn EEG – Subsets Overview

The dataset contains five physiologically distinct subsets:

Set Z

Surface EEG – Healthy subjects – Eyes open

Set O

Surface EEG – Healthy subjects – Eyes closed

Set N

Intracranial EEG – Epileptic patients – Seizure-free (hippocampus)

Set F

Intracranial EEG – Epileptic patients – Seizure-free (epileptogenic zone)

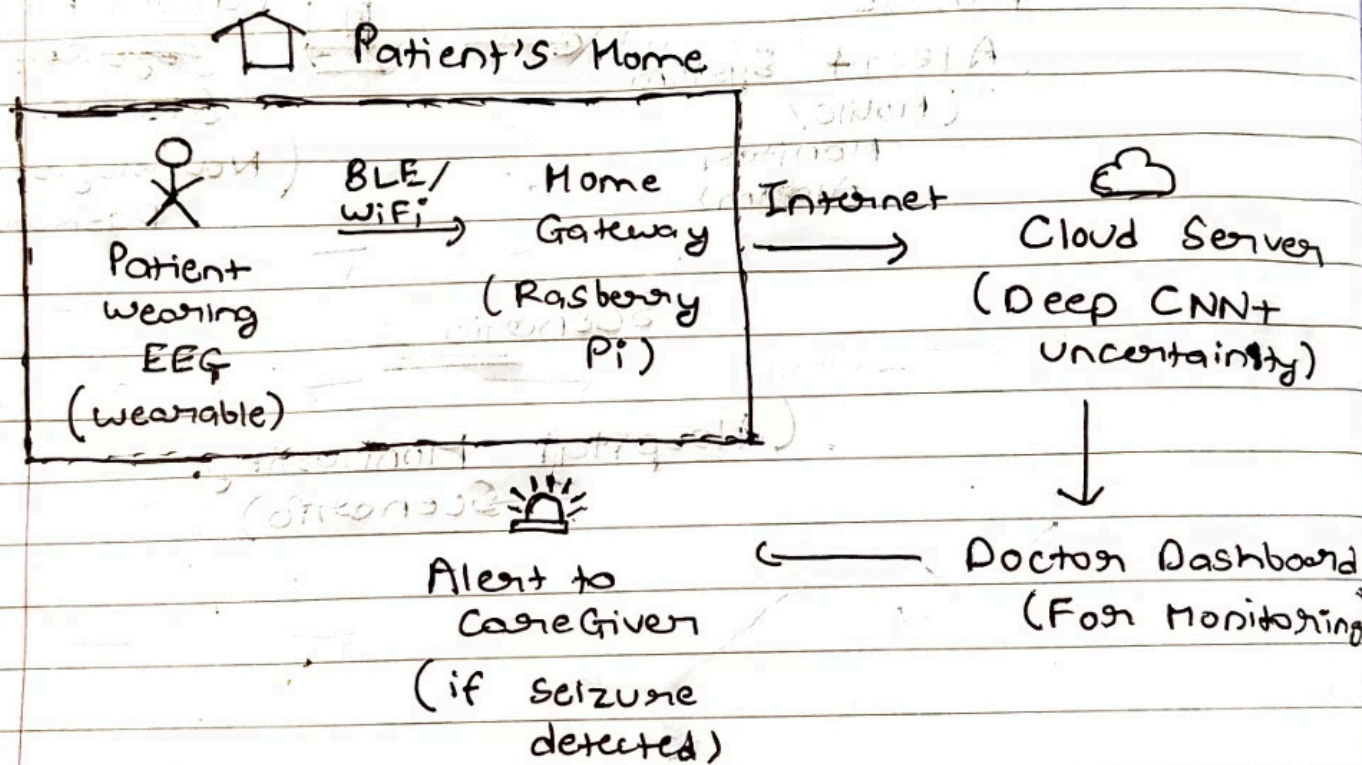
Set S

Intracranial EEG – Epileptic patients – Active seizure

Why Bonn EEG for Initial Validation?

- Lightweight and computationally efficient
- Fast CNN experimentation
- Ideal for model splitting validation
- Enables uncertainty threshold tuning
- Supports full feedback-loop simulation

The proposed framework is not tied to any specific dataset, which makes it flexible and adaptable. In future phases, it can be scaled and applied to larger clinical datasets to further validate and enhance its performance.



Scenario 1

(Home based Monitoring)

This scenario shows home-based monitoring. The wearable sends semantic tokens and transmits them to the home gateway via Bluetooth. The gateway performs basic validation, timestamp verification, and temporary caching to ensure reliable transmission. It then forwards the tokens securely to the cloud. The cloud performs analysis and uncertainty estimation. If a seizure is detected, alerts are sent to the caregiver and the doctor remotely.

Full Edge Approach

Raw EEG Signal



L1 → L2 → L3 → L4 → L5 (CNN Layers)



Seizure Prediction

(All on wearable device)

Issues with Full Edge

- Deep CNN too heavy for microcontroller
- High computation energy
- Memory constraints
- Thermal issues
- Limited model complexity
- Lower accuracy possible

Key Problem:

Edge devices cannot support full deep models efficiently

Full Cloud Approach

Raw EEG Signal



wearable device



Transmit Full RAW DATA ⇒ Cloud Server

L1 → L2 → L3 → L4 → L5



Seizure prediction

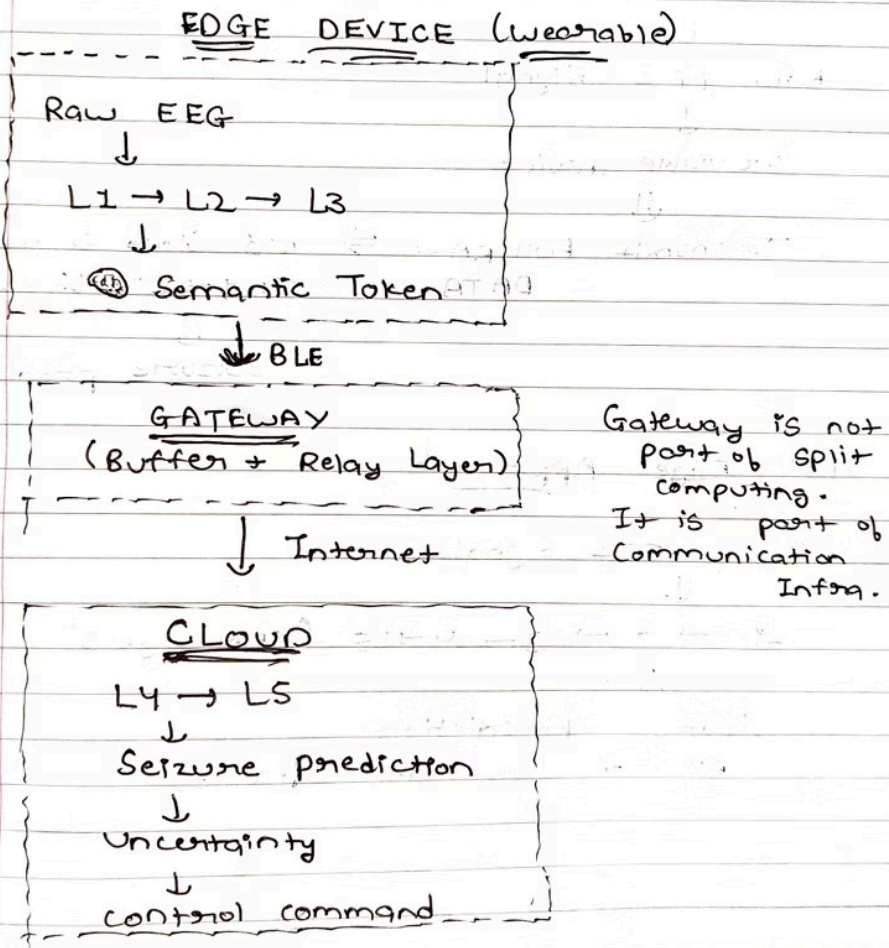
Issues with Full Cloud

- Continuous raw data transmission
- Very high communication energy
- Large bandwidth consumption
- Battery drains quickly
- Network dependency
- Latency issues

Key Problem:

Communication is the biggest energy consumer.

Split Computing



Why This Is Better

- Heavy computation offloaded to cloud
- Only compact token transmitted
- Lower communication energy
- Lower edge computation load
- Adaptive sensing possible
- Balanced energy distribution

Key Insight:

We distribute intelligence optimally across devices.

Bridging to Real-World Deployment

What Is Missing Beyond Dataset-Based Validation

Multi-Modal Sensing & Fusion

- Current system uses **single-channel EEG**
- Real deployment requires multi-sensor integration:
 - **EEG** (primary neural activity)
 - **ECG** (cardiac correlation)
 - **Accelerometer** (motion/artifact detection)
- Enables:
 - distinguishing seizures vs motion artifacts
 - reducing false positives

Requires synchronized acquisition and feature-level fusion across modalities

Communication-Aware & Robust Inference

- Real-world networks introduce:
 - packet loss
 - latency
 - intermittent connectivity

Needed:

- adaptive transmission (prioritize important tokens)
- local fallback inference when disconnected

Inference must adapt to both **data uncertainty** and **network conditions**

Personalization & Domain Shift

- EEG patterns vary across patients and sessions
- Dataset-trained models may not generalize

Challenges:

- inter-patient variability
- long-term signal drift

Requires patient-specific adaptation (fine-tuning or adaptive thresholds)

Stable Closed-Loop Control

- Current feedback: simple threshold-based switching
- Real systems must avoid:
 - frequent mode oscillations
 - unnecessary energy spikes

Needed:

- hysteresis / state-based control
- constraints on switching frequency

Converts system into a **stable feedback-controlled adaptive system**

Reliability, Safety & Real-Time Operation

- Continuous streaming with **sliding-window processing**
- Handling missing/noisy data
- Data integrity checks and secure transmission
- Fail-safe alert mechanisms for critical events

Simulation Workflow (On Local Device by simulating the setup)

Full Cloud Mode

EEG Signal → Cloud → Full CNN Inference → Prediction

Full Edge Mode

EEG Signal → Edge Device → Full CNN Inference → Prediction

Split Computing Mode

EEG Signal → Edge Encoder → Semantic Token → Cloud Decoder → Prediction

Experimental Objectives

- Compare communication bandwidth across different inference strategies
- Measure end-to-end latency
- Evaluate seizure classification accuracy
- Analyze effectiveness of semantic token transmission
- Validate feasibility of uncertainty-aware split computing

Performance Comparison Across Three Inference Schemes

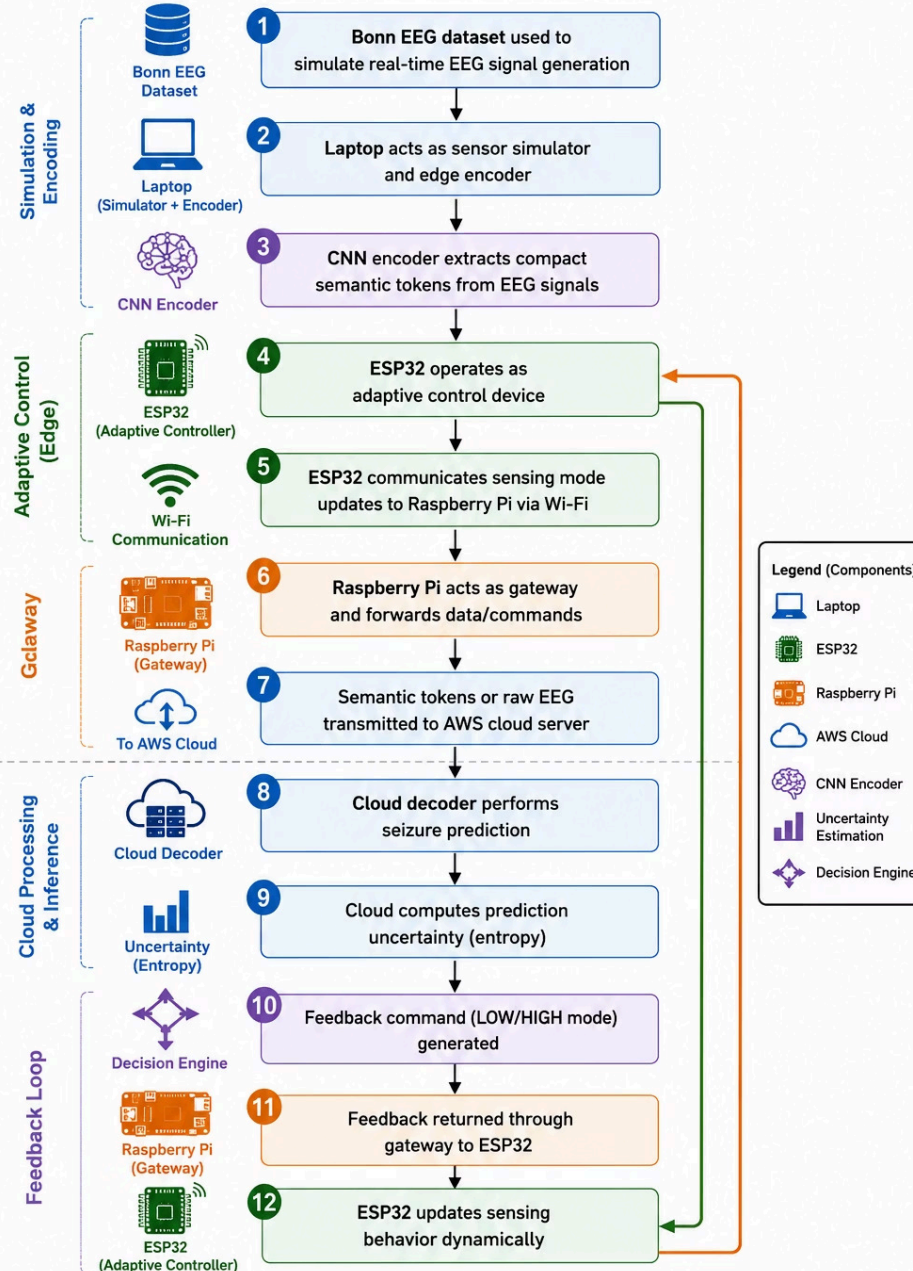
Metric	Cloud	Edge	Split
Accuracy	94.8%	94.8%	94.8%
Precision	95.12%	95.12%	95.12%
Recall	94.8%	94.8%	94.8%
F1-Score	94.49%	94.49%	94.49%
Bandwidth Usage	16003.91 KB	0.49 KB	500 KB
Latency	27.34 ms	1.17 ms	1.84 ms

NOTE: Here Cloud means complete simulation on cloud, Edge means complete simulation on Edge and Split means the proposed Split computing method

Key Observations

- Split computing maintained the same classification performance as full cloud and edge inference.
- Semantic token transmission significantly reduced bandwidth compared to raw EEG transmission.
- Split computing achieved low latency while offloading heavy computation to the cloud.
- Results demonstrate efficient communication without compromising seizure detection accuracy.

End-to-End Workflow of Actual Implementation



Hardware Components

Component	Purpose
ESP32-WROOM-32	Adaptive sensing controller and feedback device
Raspberry Pi 3B	Gateway device for communication and routing
AWS EC2 Cloud Server	Cloud inference and entropy estimation
Laptop/Desktop	Model training, simulation, and debugging
USB Serial Communication	Between Laptop and ESP32 , for uploading scripts

Software and Frameworks

Software / Framework	Usage
Python	Backend and ML implementation
PyTorch	CNN model training and inference
Flask	Cloud REST API server
NumPy	EEG signal processing
scikit-learn	Dataset split and evaluation metrics
Arduino IDE	ESP32 programming
PySerial	Serial communication
Ubuntu Server 24.04	Cloud server OS
Bonn EEG Dataset	EEG seizure detection dataset

Proposed Architecture

Practical Implementation

CNN encoder deployed on an edge gateway device for low-latency local inference

CNN encoder executed on a laptop-based edge gateway because of ESP32 computation limitations

Real-time EEG acquisition directly from wearable sensors during live monitoring

Bonn EEG dataset used to realistically simulate EEG streaming, allowing seizure-like events to be evaluated without requiring continuous hardware collection due to hardware limitations

ESP32 was used for adaptive sensing control and wireless communication with the Raspberry Pi gateway.

Laptop simulated EEG acquisition and performed edge-side CNN encoding, while the ESP32 handled transmission of sensing updates and control signals to the Raspberry Pi.

Problems Faced During Implementation

- Raspberry Pi faced power limitations and instability when multiple USB devices (keyboard, mouse, pendrive, etc.) were connected simultaneously, causing occasional reboots.
- PyTorch and deep learning framework support on Raspberry Pi 3B introduced compatibility and performance issues.
- Due to these limitations, the CNN encoder was finally executed on the laptop instead of the Raspberry Pi.
- Wireless communication setup between the ESP32 and Raspberry Pi was delayed due to Raspberry Pi Wi-Fi configuration.
- AWS cloud deployment required API testing and port configuration.
- Real-time synchronization between edge, gateway, and cloud increased overall system complexity which took some time to setup.
- Since actual EEG acquisition hardware/sensors were unavailable, the Bonn EEG dataset was used to simulate EEG signals.
- The laptop was additionally used to simulate sensor-side EEG acquisition and streaming.

Possible Solutions and Future Improvements

- Use a more powerful Raspberry Pi version or edge AI hardware (e.g., Raspberry Pi 5, Jetson Nano, Coral TPU) for efficient PyTorch/TensorFlow execution.
- Replace simulated EEG data with real EEG acquisition hardware and biomedical sensors.
- Integrate wearable EEG headsets for real-time patient monitoring.
- Optimize semantic token size using quantization and compression techniques.
- Implement secure encryption and authentication for privacy-preserving medical communication.
- Extend the framework to multi-channel EEG and larger clinical datasets for improved robustness.
- Using dynamically determined patient-specific thresholds for adaptive monitoring instead of static thresholds.